

Checkpoint F1: Vector fields

The moral of this checkpoint is that technology makes exploring vector fields very easy.

1. Use CalcPlot3D or something similar to generate a graph of the 2D vector field $\vec{F}(x,y) = \langle 2x - 3y, 4x - 2y \rangle$. Take a screenshot and put it in your submission.
2. Maybe this vector field is just a special case of a more general thing

$$\vec{F}(x,y) = \langle 2x - ay, bx - 2y \rangle.$$

Explore how this vector field looks for a couple of different values of the constants a and b . Maybe pick some where $a = b$, where $a < b$, where $a > b$. Take screenshots and include them in your submission.

3. What kinds of general things can you say about the behavior of this vector field?